6.7.1 Nutritional Needs during Pregnancy

In addition to the RDA for an adult women the nutritional needs increase during pregnancy.

Energy

Energy requirement during pregnancy is increased because of the additional energy required for growth and activity of foetus, growth of placenta and maternal tissues, increase in maternal body size and steady rise in BMR.

Protein

An additional protein intake is essential for:

- Growth of the foetus.
- Development of placenta
- Enlargement of uterus, mammary glands
- Increased maternal blood volume
- Formation of amniotic fluid

Fat

ICMR expert committee has suggested an intake of 30g of visible fat/day during pregnancy.

Calcium

Additional calcium is needed for the growth and development of bones as well as teeth of the foetus and also for the protection of calcium resources of the mother to meet the high demand of calcium during lactation.

Iron

The requirement of iron increases from 21mg/day to 35mg/day during pregnancy.

The increased requirement is due to

i. expansion of maternal tissues including red cell mass, iron content of placenta and blood loss during parturition.

ii. to build the iron store in foetal liver to last for at least 4-6 months after birth. This is because the baby's first food milk is deficient in iron.

DO YOU KNOW?

Pica refers to the compulsion for persistent ingestion of unsuitable substances that have little or no nutritional value like starch, clay and chalk.

Iodine

Due to increase in BMR, iodine needs are also enhanced during pregnancy.

Zinc

Deficiency of zinc adversely affects the outcome of pregnancy. Zinc deficiency leads to foetal mortality, foetal malformations and reduced intra uterine growth rate.

Sodium

Normal sodium intake without restriction is advised during pregnancy. Sodium is restricted when there is oedema or hypertension.

Vitamins

An additional allowance of Vitamin A is needed during pregnancy. Vitamin D is essential as it enhances maternal calcium absorption. Vitamin K is required for synthesis of prothrombin which is essential for normal coagulation of blood. A liberal vitamin K level in the mother's blood is helpful in preventing neonatal haemorrhage. Vitamin C, pyridoxine, and vitamin B12 needs are increased during pregnancy.

Folic acid

Folic acid is essential for increased blood formation i.e. haematopoiesis and for

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